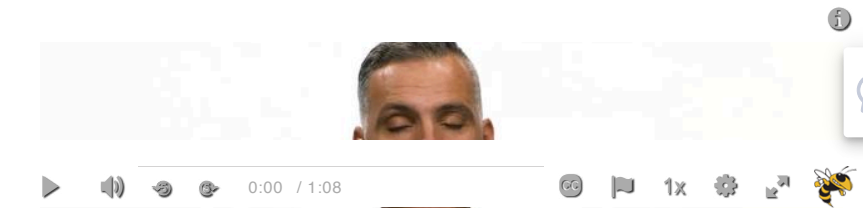


Lesson Five Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Introduce the “*network efficiency*” concept and define the relevant quantitative metrics
- Introduce the “*network clustering*” concept and define the relevant quantitative metrics
- Explain differences between “*small-world*” networks and random/regular/power-law networks
- Introduce the “network motifs” concept and define the relevant quantitative metrics
- Explain the significance of “*small-world*” networks through case studies

Required and Recommended Reading

Required Reading

- Sections 3.8, 3.9, 5.10 from A-L. Barabási, [Network Science \(http://www.networksciencebook.com/\)](http://www.networksciencebook.com/), 2015
- Sections 3.1, 3.2, 20.1, 20.2 - D. Easley and J. Kleinberg, [Networks, Crowds and Markets](https://www.cs.cornell.edu/home/kleinber/networks-book) , <https://www.cs.cornell.edu/home/kleinber/networks-book>, Cambridge Univ Press, 2010 (also available [online](https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book.pdf) , <https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book.pdf>).
- [Structure and function of the feed-forward loop network motif](https://doi.org/10.1073/pnas.2133841100%E2%80%8B%C2%A0), , <https://doi.org/10.1073/pnas.2133841100%E2%80%8B%C2%A0>, S. Mangan and U. Alon, PNAS October 14, 2003 100 (21) 11980-11985

Recommended Reading

Assignments

 Lesson-5: Knowledge Check

Lesson-5: Quiz

Module Two [Assignment]: Project Two